

# MuJoCo

control signal  $\rightarrow$  actuator  $\rightarrow$  force/torque  $\rightarrow$  physics  $\rightarrow$  new state

- Motor or actuator
- Force or torque
- Mass or inertia
- Gravity or friction
- Velocity or acceleration
- Passage of simulated time

$\rightarrow$  mujoco introduces these missing physics

$\rightarrow$  MuJoCo separates the sim.  
into two main ideas:

- Model: the mostly fixed blueprint—bodies, joints, masses, geometry, gravity, actuators, timestep.
- Data: the changing state at this moment—joint positions, velocities, forces, controls, simulation time.

qpos: joint position/config.

qvel: joint velocity

ctrl: actuator command

for a simple one-joint pendulum:

$qpos = [\theta]$      $qvel = [\dot{\theta}]$      $ctrl = [u]$

$\rightarrow$  gravity and velocity can make it move  
 $ctrl = 0 \neq$  not moving  
it means actuator not applying any command

then one sim step behaves:

$$(q_{pos_t}, q_{vel_t}, ctrl_t) \xrightarrow{\text{physics}} (q_{pos_{t+1}}, q_{vel_{t+1}}, ctrl_{t+1})$$

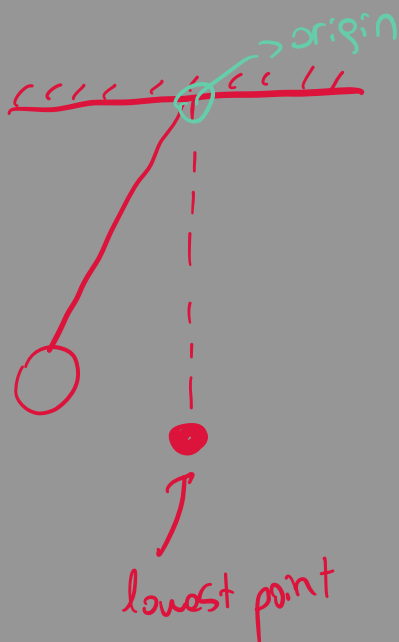
### Basic Simu. Loop:

Load model

- Create changing simulation data
- Set control command
- Advance physics by one timestep
- Read new position and velocity
- Repeat

!  $\Rightarrow$  FK: assign angles directly

MuJoCo: gravity and dynamics evolve  
angle and velocity over time



- speed is largest

becuz  $\rightarrow$  gravitational p.e  $\rightarrow$  kinetic energy  
fully

> commanding an actuator

ctrl  $\rightarrow$  torque  $\rightarrow$  angular acceleration  $\rightarrow$  qvel  $\rightarrow$  qpos

$\rightarrow$  applying positive motor command while at rest  
first velocity responds  
then position

actuator torque = gear  $\times$  ctrl

! actuator torque  $\neq$  total torque  $\rightarrow \tau_{\text{total}} = \tau_{\text{act.}} + \tau_{\text{gravity}} + \tau_{\text{damping}}$

- $\text{ctrl} = +1 \rightarrow$  torque +10, angle  $+59.220^\circ$ , velocity  $-0.455$
- $\text{ctrl} = -1 \rightarrow$  torque  $-10$ , angle  $-59.220^\circ$ , velocity  $+0.455$

The final velocity signs point back toward zero because gravity has reversed each pendulum.

1. What is the difference between `model` and `data`?
2. What do `qpos`, `qvel`, and `ctrl` represent?
3. What does `mj_step()` do?
4. Why did `ctrl=1` produce actuator torque 10?
5. Why can velocity be negative while actuator torque remains positive?
6. Does `ctrl=0` guarantee the robot stops? Why?

1. model is fixed stuff that acts like blueprint  
data is the thing we adjust and exam

2.  $qpos$  = joint pos. ;  $qvel$  = joint velocity ;  $ctrl$  = <sup>actuator</sup> command

3. advances the sim. by one timestep e.g. 0.002s

4. becuz it gets mult. by gear

5.

- Velocity describes which way it is moving now.
- Torque contributes to angular acceleration—how velocity is changing.
- Positive actuator torque does not overwrite a negative velocity.
- In our experiment, gravity produced enough opposing torque to reverse the pendulum; afterward, the motor's positive torque opposed that negative motion.

6. no, it still may be swinging because of other torques applied